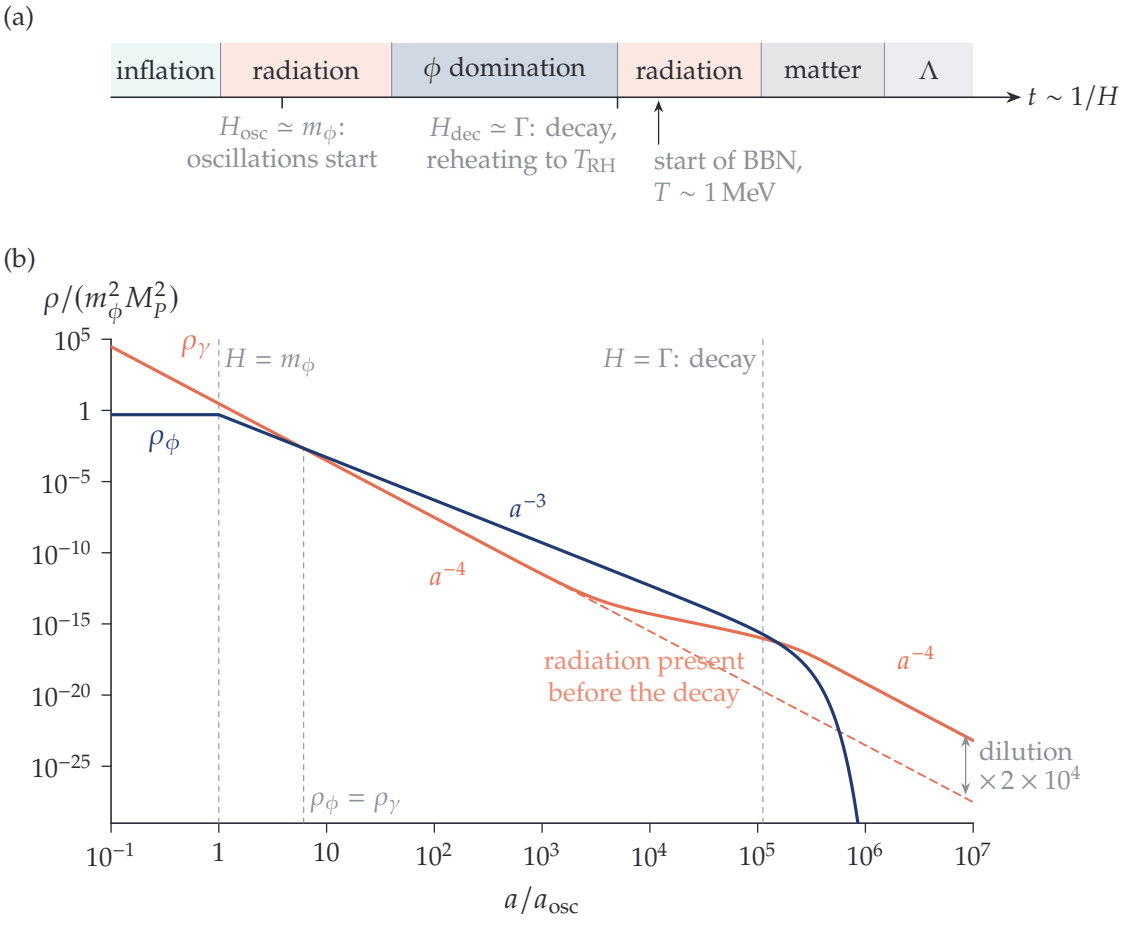

(a)
inflation
radiation
ϕ domination
radiation
matter
Λ
$t \sim 1/H$
$H_{\rm osc} \simeq m_\phi$: oscillations start
$H_{\rm dec} \simeq \Gamma$: decay, reheating to $T_{\rm RH}$
start of BBN, $T \sim 1\,{\rm MeV}$
(b)
$\rho/(m_\phi^2 M_P^2)$
10^5
1
10^{-5}
10^{-10}
10^{-15}
10^{-20}
10^{-25}
ρ_γ
ρ_ϕ
$H = m_\phi$
$H = \Gamma$: decay
a^{-3}
a^{-4}
radiation present before the decay
a^{-4}
dilution $\times 2 \times 10^4$
$\rho_\phi = \rho_\gamma$
10^{-1}
1
10
10^2
10^3
10^4
10^5
10^6
10^7
$a/a_{\rm osc}$